

1. Administrative & Study Identification

Full Study Title: A Phase 2, Multi-center, Open-label, Randomized Study of Oral Asciminib Added to Imatinib Versus Continued Imatinib Versus Switch to Nilotinib in Patients With CML-CP Who Have Been Previously Treated With Imatinib and Have Not Achieved Deep Molecular Response **Short Title/Acronym:** ASC4MORE (Study of Efficacy and Safety of Asciminib in Combination With Imatinib in CML-CP) **Protocol Number:** CABL001E2201 **NCT Number:** NCT03578367 **Posting ID:** 590946442528261238 **Sponsor/Company Name:** Novartis Pharmaceuticals **Investigational Product:** Asciminib (ABL001), Imatinib, Nilotinib **Phase of Development:** Phase 2 **Trial Status:** Completed **Medical Monitor:** Novartis Pharmaceuticals

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess whether asciminib (40 mg or 60 mg) once daily in combination with imatinib 400 mg once daily is more effective than continued imatinib alone by testing the Deep Molecular Response (\$MR^{4.5}\$) rate at 48 weeks. **Secondary Objectives:** To evaluate the \$MR^{4.5}\$ rate at 48 weeks in the asciminib add-on arms compared to the nilotinib switch arm; to evaluate the best observed \$MR^{4.5}\$ rate up to 48 weeks; and to assess the \$MR^{4.5}\$ rate sustained up to 96 weeks, alongside overall safety and tolerability.

2.2 Background & Rationale To address the gap in achieving Deep Molecular Response (DMR) in chronic myeloid leukemia in chronic phase (CML-CP) patients who have been treated with frontline imatinib for at least one year but have not reached \$MR^{4.5}\$. Preclinical data suggests that combining asciminib—a BCR-ABL1 inhibitor that Specifically Targets the ABL Myristoyl Pocket (STAMP)—with an ATP-competitive TKI like imatinib can provide more potent BCR-ABL1 inhibition and prevent the emergence of resistance mutations, offering a pathway to deeper and more durable molecular responses.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Continued imatinib alone or switch to nilotinib) **Blinding:** Open-label **Randomization:** Randomized, Parallel Assignment

3.2 Planned Enrollment & Duration Target **N**: 104 participants
Number of Sites: Multi-center (Global: including Asia, Australia, Europe, North America, South America) **Estimated Study Start:** 22-Nov-2018 **Estimated Primary Completion:** 08-Nov-2021

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female patients ≥ 18 years of age. 2. Confirmed diagnosis of Philadelphia chromosome-positive Chronic Myeloid Leukemia in chronic phase (CML-CP). 3. Minimum of one year (12 calendar months) treatment with imatinib first line for CML-CP (patients have to be on imatinib 300mg or 400 mg QD at randomization). *For Korea only: (i) a minimum of one year of prior treatment with imatinib for patients with BCR-ABL levels $> 0.1\%$, $\leq 1\%$ IS at randomization; (ii) a minimum of two years of prior treatment with imatinib for patients with BCR-ABL levels $> 0.01\%$, $\leq 0.1\%$ IS at randomization.* 4. BCR-ABL1 levels $> 0.01\%$ IS (International Scale) and $\leq 1\%$ IS at the time of randomization confirmed with central assessment; no prior achievement of deep molecular response (MR^4 IS) confirmed by 2 consecutive tests at any time during prior imatinib treatment (an isolated single test $< 0.01\%$ is allowed if not within 9 months prior). 5. Adequate laboratory values before randomization: Absolute Neutrophil Count $\geq 1.5 \times 10^9/L$, Platelets $\geq 75 \times 10^9/L$, Hemoglobin ≥ 9 g/dL, Serum creatinine < 1.5 mg/dL, Total bilirubin $\leq 1.5 \times$ ULN (Gilbert's syndrome $\leq 3.0 \times$ ULN), AST $\leq 3.0 \times$ ULN, ALT $\leq 3.0 \times$ ULN, Alkaline phosphatase $\leq 2.5 \times$ ULN, Serum lipase $\leq 1.5 \times$ ULN. (Increases in potassium up to 6.0 mmol/L, calcium up to 12.5 mg/dL, and magnesium up to 3.0 mg/dL are acceptable if associated with normal creatinine clearance).

4.2 Exclusion Criteria 1. Treatment failure according to European LeukemiaNet (ELN) 2013 criteria during prior imatinib treatment. 2. Known second chronic phase of CML after previous progression to Accelerated Phase (AP) or Blast Crisis (BC). 3. Previous treatment with any tyrosine kinase inhibitors (TKIs) other than imatinib. 4. History of ECG abnormalities indicating significant risk (e.g., QTcF ≥ 450 msec for males or ≥ 460 msec for females, history of myocardial infarction, angina pectoris, or CABG within 6 months, long QT syndrome). 5. Severe and/or uncontrolled concurrent medical disease causing unacceptable safety risks or compliance compromise (e.g., uncontrolled diabetes, active or uncontrolled infection, uncontrolled clinically significant hyperlipidemia, and high serum amylase). 6. History of acute pancreatitis within 1 year prior to randomization or medical history of chronic pancreatitis; ongoing acute or chronic liver disease. 7. History of other active malignancy within 3 years prior to randomization with the exception of basal cell skin cancer, indolent prostate cancer, and carcinoma in situ treated curatively.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * Arm 1/2: Asciminib (40 mg or 60 mg) once daily + Imatinib (Trade Name: Gleevec, ATC Code: L01EA01) 400 mg once daily. * Arm 3: Continued Imatinib 400 mg once daily. * Arm 4: Switch to Nilotinib 300 mg twice a day (BID). * Arm 5: Asciminib 80 mg once daily as a single agent. **Route of Administration:** Oral (Film-Coated tablets) **Duration of Treatment:** Up to 96 weeks (with an option for imatinib continuation patients not achieving $SMR^{4.5}$ at 48 weeks to cross over within 4 weeks to the asciminib 60 mg add-on arm).

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care adjusting for electrolyte imbalances. **Prohibited:** Concomitant medications with a "known" risk of Torsades de Pointes or that strongly interact with the QTcF interval.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, ECG, Molecular labs (BCR-ABL1), Chemistry
Baseline	Day 0	Randomization, First Dose
Interim 1	Week 24 $\pm$ 7	Safety Labs, Efficacy Assessment (Interim molecular response)
Interim 2	Week 48 $\pm$ 7	Primary Efficacy Assessment ($SMR^{4.5}$ evaluation), Cross-over eligibility assessment
End of Study	Week 96	Final Efficacy/Safety Labs, IP Accountability, Transition to follow-up

(Note: A safety follow-up period continues for 30 days after the last dose received)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Proportion (Percentage) of participants still treated with the randomized treatment at 48 weeks and who are in $SMR^{4.5}$ (BCR::ABL1 ratio of $\leq 0.0032\%$ on the International Scale) at 48 weeks ($\pm$ assessment window) in the asciminib add-on arms versus the continued imatinib arm. **Measurement Method:** Polymerase chain reaction (PCR) analysis of blood samples (Central Laboratory).

7.2 Secondary & Exploratory Endpoints * Rate of $SMR^{4.5}$ at 48 weeks in asciminib add-on arms versus the nilotinib arm. * Rate of $SMR^{4.5}$ by 48 weeks (Best observed rate). * Rate of $SMR^{4.5}$ at 96 weeks.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of AEs via clinical evaluation, lab tests (including pancreatic enzymes, lipid profiles, and liver function tests), and ECG monitoring.

Serious Adverse Events (SAEs): Reported as per standard regulatory requirements within 24 hours. **Data Monitoring**

Committee (DMC): Regular safety monitoring reviews conducted by the sponsor to evaluate the benefit/risk balance, including a pre-planned interim analysis after 50% enrollment.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) Population (all randomized participants for primary efficacy endpoints); Safety Set (all patients receiving at least one dose of study medication).

Sample Size Justification: Target enrollment of $N=104$ designed to provide sufficient statistical power to compare the Deep Molecular Response ($MR^{4.5}$) rate between the combined asciminib + imatinib arms and the imatinib alone arm at 48 weeks. **Handling of Missing Data:** Dropouts or individuals discontinuing the randomized treatment prior to week 48 are treated as non-responders for the primary endpoint analysis.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine on-site and remote monitoring visits to ensure protocol adherence, drug accountability, and valid BCR-ABL1 sample collection. **Audit Trail:** Continuous centralized data checking and eCRF queries executed by Novartis site management.

11. Study Identifiers & Governance

Primary ID: CABL001E2201 **Registry Number:** NCT03578367 **Posting ID:** 590946442528261238 **Sponsor/Lead Organization:** Novartis

Pharmaceuticals Study Title: Study of Efficacy and Safety of Asciminib in Combination With Imatinib in Patients With Chronic Myeloid Leukemia in Chronic Phase (CML-CP) **Official Regulatory Title:** A Phase 2, Multi-center, Open-label, Randomized Study of Oral Asciminib Added to Imatinib Versus Continued Imatinib Versus Switch to Nilotinib in Patients With CML-CP Who Have Been Previously Treated With Imatinib and Have Not Achieved Deep Molecular Response

12. Clinical Framework (CML Specific)

Primary Condition: Chronic Myeloid Leukemia in Chronic Phase (CML-CP) / Preferred UMLS Name: Chronic Myeloid Leukemia **Disease Areas:** CML, Chronic Myelogenous Leukemia, Leukemia, Myeloid Chronic, Hematologic Diseases (Preferred Name: Hematological Disease). **Semantic Type / Definition:** Disease or Syndrome / Disorders of the blood and blood forming tissues. **Terminology Codes:** MeSH: D006402 | SNOMED: 34093004 | HPO: HP:0001871 **Diagnosis Threshold:** BCR-ABL1 > 0.01% to $\leq 1\%$ (IS) without achieving prior $MR^{4.5}$ **Medical Methodology:** Tyrosine Kinase Inhibitor (TKI) Therapy **Optimization Target:** Achievement of Deep Molecular Response ($MR^{4.5}$ / BCR-ABL1 ratio $\leq 0.0032\%$)

13. Study Procedures & Methodology

Management Pathway: Step-up TKI Add-on vs. Switch Therapy **Education Protocol (HCP):** Investigator protocol training on dose modifications, strict ECG guidelines, and cross-over criteria. **Education Protocol (Patient):** Strict oral compliance requirements and TKI side-effect management. **Quality Audit Mechanism:** Centralized BCR-ABL1 PCR laboratory processing to ensure $MR^{4.5}$ accuracy.

14. Observation & Follow-Up Schedule

Total Duration: Up to 96 weeks of treatment + 30 days safety follow-up **Onsite Visit Cadence:** Minimum specific visits at Weeks 24, 48, and 96 **Remote Monitoring Frequency:** Supplemental lab evaluations per local standards **Checklist Review Interval:** Continuous throughout the 96-week treatment cycle

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				